FRQ 2 Practice – Lawn Be Gone

The company "Lawn Be Gone" sells pine straw as an alternative for homeowners wishing to replace grass lawns. On January 1st of 2020 ($t = 0$), the company had sold 5000 bales of pine straw in the past year. On January 1st of 2026 ($t = 6$), the company had sold $24{,}000$ bales of pine straw in the past year.

The amount of pine straw sold by this company can be modeled by the exponential function S given by $S(t) = a(b)^t$, where $S(t)$ is the number of bales sold in the year preceding time t, and t is the number of years since January 1st of 2020.

A.

(i) Use the given data to write two equations that can be used to find the values of a and b in the expression for $S(t)$.

(ii) Find the values for a and b as decimal approximations.

B.

(i) Use the given data to find the average rate of change of the quantity sold, in bales per year, from $t = 0$ to $t = 6$ years. Express your answer as a decimal approximation. Show the computations that lead to your answer.

(ii) Use the average rate of change found in (i) to estimate the quantity of pine straw sold, in bales, for $t = 4$. Show the work that leads to your answer.

C. Initially, Lawn Be Gone only sells their product at local stores. Once Lawn Be Gone's yearly sales reach $40{,}000$ bales, they plan to begin selling online and shipping to a wider audience. This will dramatically alter the quantity they sell. Use model S to respond to the following.

(i) Write and solve an equation whose solution gives the time t, in years since January 1st of 2010, when the yearly sales by Lawn Be Gone reaches $40{,}000$ bales. Express your value for t as a decimal approximation.

(ii) Explain how the value for t found in part C(i) can be used to determine an appropriate domain restriction for the model S. Refer to the context of the question in your response.